

Homework 1(g) - Solutions

MT2005 - Probability and Statistics

February 24, 2025

1 Z-Scores

1.1 Calculation of Z-Scores

The z-score is calculated as:

$$Z = \frac{x - \mu}{\sigma} \quad (1)$$

where μ is the mean and σ is the standard deviation.

Given the dataset:

Employee	Salary (Rupees)
1	55000
2	65000
3	70000
4	75000
5	90000
6	150000

We first compute:

$$\mu = \frac{55000 + 65000 + 70000 + 75000 + 90000 + 150000}{6} = 84250$$

$$\sigma = \sqrt{\frac{(55000 - 84250)^2 + \dots + (150000 - 84250)^2}{6}}$$

Substituting values, we obtain the z-scores:

Salary (Rupees)	Z-Score
55000	Z_1
65000	Z_2
70000	Z_3
75000	Z_4
90000	Z_5
150000	Z_6

An outlier is identified if $|Z| > 3$.

2 Percentiles

2.1 10th and 90th Percentiles

Percentiles are computed using:

$$P_k = \frac{k}{100}(n+1) \quad (2)$$

Substituting values, we determine P_{10} and P_{90} .

3 Percentile Rank

3.1 Percentile Rank of 75000

$$\text{Percentile Rank} = \frac{\text{Number of values} < 75000}{n} \times 100 \quad (3)$$

4 Deciles

4.1 4th and 7th Deciles

Using:

$$D_k = \frac{k}{10}(n+1) \quad (4)$$

we compute D_4 and D_7 .

5 Quartiles

5.1 1st, 2nd, and 3rd Quartiles

Using:

$$Q_k = \frac{k}{4}(n+1) \quad (5)$$

we compute Q_1 , Q_2 , and Q_3 .

6 Outliers using IQR

6.1 IQR Method

$$\text{IQR} = Q_3 - Q_1 \quad (6)$$

Outliers satisfy:

$$x < Q_1 - 1.5 \times \text{IQR} \quad \text{or} \quad x > Q_3 + 1.5 \times \text{IQR} \quad (7)$$

7 Skewness and Kurtosis

Skewness and kurtosis are computed using:

$$\text{Skewness} = \frac{\sum (x - \mu)^3}{n\sigma^3} \quad \text{Kurtosis} = \frac{\sum (x - \mu)^4}{n\sigma^4} - 3$$

8 Boxplot

Python code for the boxplot:

```
import matplotlib.pyplot as plt import seaborn as sns salaries = [55000, 65000, 70000, 75000]
```